

	Shah Zain: data and evaluation	Ali Usama: geometry
Data needed	Scans, labels, calibration, object database	Scans only
Session 1	Load scans, calibration and labels	Load and display a raw scan
	Convert label boxes into LiDAR coordinates	Crop the scan to the area of interest
	Check boxes sit on the cars in 10 frames	Downsample points with a voxel grid (own code)
	Find which points are inside a box	3D viewer with points coloured by height
		Draw boxes in the viewer (test with a fake box)
Session 1 done when	Boxes match the cars in 10 frames	A downsampled scan shows in 3D with a box
Session 2	Box overlap (IoU) measure, with tests	MSAC ground plane fitting (own code)
	Fixed train/validation split	Compare with Open3D's RANSAC on the same scans
	Count objects per class (cars, pedestrians, cyclists)	Measure speed on 10 scans
	Compute features per object (size, height, number of points, distance, shape) and save them	Show ground points in grey
		Note scans where one plane fails (slopes)
Session 2 done when	Overlap tests pass, feature dataset saved	Ground removed correctly on 10 scans